

National Policy and Legislative Framework for AI
Governance in Law Enforcement Agencies and
Criminal Justice System of Pakistan



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Abstract

Artificial Intelligence (AI) is rapidly transforming all sectors of the society into a new technological world. Specifically speaking, the law enforcement practices have been transformed by AI offering enhanced surveillance, data analytics, and identification capabilities across the globe. Some countries like China and USA have adopted this advancement through their national policies on AI governance. In Pakistan, however, this integration of AI tools such as facial recognition, fingerprint matching, data protection protocols, and predictive policing is underway, yet these developments are occurring in a legal vacuum. This paper will be focusing on the application of AI in Criminal Justice System and Law Enforcement Agencies of Pakistan. The Criminal Justice System (CJS) requires more transparency and accountability which can only be assured by enhancing the mechanism of law enforcement agencies in Pakistan. AI can be more beneficial in Criminal Justice System decreasing procedural complexities, culprits' smooth identification, and victim's easy access to justice. However, among law enforcement agencies, the Police Department has been mainly focused.

This paper will elaborate how AI can play an active role in enhancement of Criminal Justice System and law enforcement agencies of Pakistan, while taking into account the shortcomings of this AI regulation. The policy for AI governance have also been proposed in this paper which would emphasize on AI governance in functioning of law enforcement agencies which includes the registration of FIR in police station, offender's prosecution, and the enforcement of court's orders.

Relaying on international best practices, and Pakistan's socio-political environment, this draft aims to ensure accountability, transparency, and constitutional compliance in AI based law enforcement practise. The proposed framework aligns with the fundamental rights provided in the Constitution of Pakistan, such as right to privacy and fair trial.

Key Words: *AI governance, legislative framework, Criminal Justice System, national policy, law enforcement agencies.*

Preamble

This policy for AI governance in law enforcement agencies and criminal justice system provides for the smooth functioning of the institutions, decreasing procedural complexities in the agencies, the identification mechanisms for accused, the data protection protocols, transparency and accountability. It also provides for the good governance of the law

enforcement agencies by eliminating administrative defects with the application of AI tools within the institutions. The loopholes in communication between citizens and the agencies have also been covered under the policy. It further deals with the regulatory framework for AI governance in simultaneous functioning of all pillars of criminal justice system dealing with the mechanisms of agencies, the prosecution, and the trial proceedings in court.

This policy is in compliance with the provisions of the Constitution of Pakistan, the Code of Criminal Procedure, the Police Rules 1934, and the Police Order 2002, ensuring the sanctity of criminal justice system of Pakistan.

Goals

This policy is proposed to introduce AI in criminal justice system of Pakistan and to enhance the performance of law enforcement agencies, especially the Police Department.

Objectives

The objectives of the Policy are;

- (a) To minimize procedural complexities in Criminal Justice System by the application of AI in law enforcement agencies.
- (b) To provide easy and smooth access of law enforcement agencies to general public.
- (c) To ensure accountability and transparency in the law enforcement agencies.
- (d) To introduce AI identification tools such as facial recognition, fingerprint registration, and data protection protocols.
- (e) To fill the gap between the law enforcement agencies and the citizens.
- (f) To regulate AI governance through legal force.
- (g) To allocate Special National AI Fund
- (h) To establish the AI Governance Committees at district levels which shall supervise the implementation of AI in law enforcement agencies.
- (i) To establish AI Governance Wing in each province and in federal territory.

AI in Law Enforcement: Global Trends and Local Adoption

Global Trends

Following countries have adopted AI;

1. USA

In the United States, the Department of Transportation is focused on enhancing public safety by researching, developing, and testing automated systems for detecting traffic accidents through video analysis. These efforts aim to ensure safer and more efficient commuter traffic across diverse environments, including varying weather, lighting, and traffic conditions. The National Artificial Intelligence Research and Development Strategic Plan highlights the challenges of analysing vast amounts of potentially relevant video and image data accurately and quickly, noting that this process can be time-consuming and susceptible to human error due to fatigue and other limitations. Additionally, in the field of fraud detection, companies such as PayPal utilize large datasets to continuously train their algorithms. This allows them to identify unusual patterns and adapt to new fraudulent behaviours, thereby staying ahead of potential threats.¹ (Riango n.d.)

2. Argentina

In Argentina, the integration of Artificial Intelligence into government operations has expanded, particularly within the realm of national security. The country's Ministry of Security has formed a specialized division known as the Applied Artificial Intelligence to Security Unit (UIAAS), tasked with addressing cybercrime. This initiative, introduced under the leadership of President Javier Milei, has prompted criticism and concern from human rights advocates and privacy specialists regarding its potential implications.² (Argentina Ministry of Security n.d.)

3. United Kingdom

A prominent example of Artificial Intelligence being utilized in law enforcement is predictive policing. This approach involves the use of AI algorithms to examine vast amounts of data—including past crime records, social media content, and environmental indicators—to forecast potential criminal activity in specific areas and timeframes. These forecasts allow law enforcement agencies to deploy personnel more strategically, increasing surveillance in zones deemed high-risk and potentially deterring criminal acts before they occur. In 2013, Kent Police became the first force in England and

¹ <https://search.app/91qBFS7XtnY73upX9>

² <https://en.cryptonomist.ch/2024/09/10/argentina-the-ministry-of-security-adopts-ai-to-combat-crime/#:~:text=The%20Ministry%20of%20Security%20of,human%20rights%20and%20privacy%20experts>

Wales to implement such a system, following a four-month pilot program in Medway that reportedly resulted in a 6% decrease in street violence. Nonetheless, the initiative was discontinued in November 2018, with Superintendent John Phillips stating that it was difficult to conclusively prove the system's effectiveness in crime reduction. Additionally, the United Kingdom has adopted various AI-powered tools across law enforcement, including facial recognition systems, natural language processing for investigative work, AI-assisted call handling, and automated tools for detecting and investigating criminal behaviour.³ (Devonshire n.d.)

4. Japan

Japan has emerged as a global leader in the adoption and advancement of Artificial Intelligence technologies. The country's Cabinet Office introduced its initial AI strategy in 2016, which has since undergone revisions, culminating in the current version known as AI Strategy 2022. In the field of public safety, AI is increasingly being employed for crime prevention and patrol operations. One such innovation is *Crime Nabi*, a user-friendly AI-driven tool developed by Singular Perturbations Inc., which has been implemented in both Japan and Brazil to support law enforcement in monitoring and preventing criminal activity.⁴ (The News on Sunday n.d.)

Adoption of AI in Pakistan

In Pakistan, though AI has been introduced in many sectors of the society, yet this application has to be governed, regulated and documented. For AI to truly thrive in Pakistan's landscape, several key challenges are must to be addressed. Many sectors and stakeholders have potential to implement AI tools in their mechanisms. Specifically, in Criminal Justice System and Law Enforcement Agencies of Pakistan, the applications of AI are to be regulated and governed. The Ministry of Planning, Development, and Special Initiatives (MoPDSI), under the leadership of Minister Ahsan Iqbal, recently held an important session of the National AI

³ <https://emergencyservicetimes.com/2024/09/27/how-artificial-intelligence-use-in-policing-is-transforming-law-enforcement/>

⁴ <https://www.thenews.com.pk/tns/detail/1303053-japans-ai-lead#:~:text=For%20public%20security%2C%20crime%20prevention,a%20victim%20of%20pick%2Dpocketing.>

Taskforce. This gathering included senior government officials, industry experts, and key stakeholders, who engaged in discussions on the AI Implementation Roadmap. The roadmap has been carefully formulated through a collaborative partnership between the National Centre of Artificial Intelligence (NCAI) and the taskforce. This strategic plan aims to boost national productivity, encourage technological innovation, and reinforce Pakistan's competitiveness in the international landscape.⁵ (Ministry of Planning Development and Special Initiative n.d.) Despite advancements in various sectors, the application and regulation of Artificial Intelligence in law enforcement remain in the early stages. In Punjab, however, some progress has been made through the implementation of the Face Trace System (FTS). Among the technologies integrated within this system is the Automated Number Plate Recognition (ANPR) System, which utilizes AI-driven tools and advanced imaging techniques to detect vehicles linked to criminal activity. This system functions by capturing and recording license plate data for further analysis and enforcement action.⁶ (Arin n.d.) However, this AI application is limited to the Province of Punjab. In other provinces of Pakistan, there is lack of AI implementation due to several challenges, the most important of which is the poor AI infrastructure.

Statistics and Challenges to the Agencies in Pakistan

The law enforcement agencies of Pakistan face various critical administrative and operational challenges in their functioning which hinders the proper administration of justice. Among various law enforcement agencies, the Police department is the one which plays crucial role in dealing with the daily crimes. The Police is the first instance of reporting the crime. Following data analytics shows the hurdles and administrative challenges to the department of Police in its proper functioning.

Police Workforce Overview

1. Islamabad Capital Territory Police (ICT Police)

The Islamabad Police operates under the oversight of the Central Police Office (CPO), which is structured into five key divisions: Operations, Security, Law & Order, Safe City, and Logistics. The total authorized personnel strength of the

⁵ Ministry of Planning Developments and Special Initiatives, Government of Pakistan (https://pc.gov.pk/web/press/get_press/1441)

⁶ Ghulam Shabir Arain, "Pakistan Executes AI-powered Criminal Identification System" Biometric Update, 2024.

Islamabad Capital Territory (ICT) Police stands at 12,625.⁷ (Islamabad Police-ICT n.d.)

2. Khyber Pakhtunkhwa Police

The Khyber Pakhtunkhwa Police currently comprises a force of 70,000 personnel, responsible for maintaining law and order in a population of 26 million. The headquarters of the KP Police, known as the Central Police Office (CPO), is located in Peshawar.⁸ (Kpk Police n.d.)

3. Punjab Police

As the largest province in Pakistan, Punjab is home to the country's most extensive police force, consisting of 179,921 officers. This force includes nine Regional Police Officers, four City Police Officers, one Capital City Police Officer, and thirty-five District Police Officers.⁹ (Punjab Police Annual-administartion-report n.d.)

Crime Statistics

Following table shows the crime statistics of 2025 in Punjab¹⁰ (Punjab Police Crimestatistics n.d.);

Cases Registered	102814
Challaned	37278
Under Investigation	61309
Cancelled	4227

Following table shows the reported crimes of 2019 in all the provinces¹¹ (National Police Bureau, Ministry of Interior n.d.);

Provinces / Federal Territory	Reported Crimes
Sindh	87162
Balochistan	9462
Khyber Pakhtunkhwa	178109
Punjab	490313

⁷ <https://islamabadpolice.gov.pk/history.php>

⁸ <https://police.kp.gov.pk/#sthash.Gtol0uml.dpuf>

⁹ <https://www.punjabpolice.gov.pk/system/files/Annual-administration-Report2014-15.pdf>

¹⁰ <https://punjabpolice.gov.pk/crimestatistics>

¹¹ National Police Bureau, Ministry of Interior.
https://www.pbs.gov.pk/sites/default/files/tables/social_statistics/crimes_reported_by_type_and_provinces.pdf

Structural and Administrative Challenges

1. Manual Record-Keeping

Currently, there is no computerized database system in place for managing police records in Pakistan. Instead, various manual, paper-based registers are used to maintain these records. While the specific names of the registers may vary by location, their functions remain consistent across police stations. At present, police stations still rely on traditional paper methods for registering First Information Reports (FIRs), which necessitate additional effort to maintain criminal records. Furthermore, tracing an individual's record is a time-consuming process that could be streamlined through the digitization of police station records.¹² (Journal of Scientific Research and Advances, Vol. 2, No. 4 n.d.)

2. FIR Registration Barriers

The initial step in reporting a crime is the registration of the First Information Report (FIR). However, the FIR registration process in Pakistan's Criminal Justice System faces numerous challenges, including delays, arbitrary practices, demands for bribes by police officers, and widespread corruption. Once an FIR is filed, the designated investigating officer is required to present the results of their investigation to the court, in accordance with Section 173 of the Criminal Procedure Code (CrPC), through the prosecution department, within a 14-day period.¹³ (Code of Criminal Procedure, 1898, s 154. n.d.). In criminal court proceedings, the report is referred to as a "challan," but it is officially known as the "Report of Police" under the Criminal Procedure Code (CrPC). For trial proceedings to begin, a police report is essential, as it provides the court with the findings of the initial investigation, enabling the timely resolution of pending cases. Unfortunately, in practice, police often take months or even years to submit the challan to the court. It has been observed that, in many instances, the police show little interest in conducting a transparent

¹² Journal of Scientific Research and Advances, Vol. 2, No. 4, 2015, 144-150.

¹³ Code of Criminal Procedure, 1898, s 154.

investigation within the required timeframe for registered cases.¹⁴ (Sandhu 2016) The Lahore police have been accused of significantly manipulating crime statistics by refusing to register First Information Reports (FIRs).¹⁵ (Dawn News n.d.)

3. Corruption and Discretion

Investigation plays a crucial role in the criminal justice system, as judges rely on police investigations to form their conclusions. Unfortunately, bribery within the police investigation process has become a common practice, often used to favour certain individuals. Furthermore, the Police Department is frequently hesitant to take disciplinary action against its own officers.¹⁶ (Sandhu 2016) According to the National Corruption Perceptions Survey 2023 released by Transparency International Pakistan (TIP), the police department continues to be perceived as the most corrupt institution, with the judiciary ranking as the third most corrupt in the country.¹⁷ (Transparency International Pakistan, “police judiciary among top corrupt sectors in Pakistan” 2013) In Punjab, citizens typically paid the highest bribes to receive police services (Rs21,186).¹⁸ (Ibid. n.d.) The TIP report indicates that the police continue to be the most corrupt sector, accounting for 30%, followed by tendering and contracting at 16%, with the judiciary ranked as the third most corrupt at 13%.¹⁹ (Ibid n.d.)

4. Delays in Investigation & Prosecution

One of the major obstacles to the effective administration of criminal justice in Pakistan is the delays in both police investigations and court trial proceedings. Substandard investigations often lead to delays in the completion of trials. Once a case is registered, the police are obligated to investigate if the offense is cognizable. However, individuals are sometimes arrested and detained by the police for extended periods without proper documentation of their arrest. Police authorities occasionally misuse their power to serve the interests of influential

¹⁴ <https://www.pakistantoday.com.pk/2023/10/16/police-criminal-trial-and-miscarriage-of-justice/>

¹⁵ <https://www.dawn.com/news/1541189?utm>

¹⁶ <https://www.pakistantoday.com.pk/2023/10/16/police-criminal-trial-and-miscarriage-of-justice/>

¹⁷ Transparency International Pakistan, “police judiciary among top corrupt sectors in Pakistan”, 2013.

¹⁸ Ibid.

¹⁹ Ibid.

political figures or settle personal vendettas. Those detained or arrested unlawfully are often held in locations other than police stations, undermining the proper legal procedures. Additionally, investigating officers have been known to fabricate recoveries, often creating fake recovery memos to support their cases.²⁰ (Hussain n.d.) Another significant cause of delays in the criminal justice system is the underperformance of the Prosecution Department. One of the primary factors contributing to delays in criminal trials is the failure of the investigation agency to submit the report under Section 173 of the Criminal Procedure Code (CrPC) or the challan in a timely manner. While it is the prosecutor's responsibility to review the report or challan before submitting it to the court, prosecutors often raise excessive technical objections, sometimes to extract bribes from the investigating officers. This practice results in further delays in the submission of challans.²¹ (Ibid n.d.)

Proposed Policy Guidelines

To enhance the Criminal Justice System and the functioning of law enforcement agencies, it is crucial to apply AI governed administrative model. This AI governed model will ensure transparency, accountability, and efficiency in the said sector.

1. AI Reception Desk

- An AI-powered board or virtual assistant, via screen or voice, receives visitors at police station.
- It interacts in Urdu, English, or any other regional languages, identifies the purpose of visit and directs them accordingly.
- Uses facial recognition or National Identity Card Number recognition and trace visitor with previous criminal record.

2. Smart FIR Counter

- AI-enabled FIR desks can auto generate FIR drafts based on victim's narration using Natural Language Processing (NLP).

²⁰ : Hussain, A Akhtar, & Hassan, "Studying the Causes of Delay in Criminal Trials under the Criminal Justice System of Pakistan" Global Sociological Review, Vol (II), 52-58.

²¹ Ibid.

- Pre checks for essential legal elements of each complaint as per the laws; CrPC, PPC, etc.
- Assigns a case number, officer-in-charge, and syncs with digital police records.

3. Real-Time Notification System

- AI system alerts officers and supervisors through immediate notification regarding;
- Delays in investigation or incomplete tasks.
- New FIRs registered in their jurisdiction.
- Status of the case or investigation.

4. AI-Controlled Police Registers

- Replace manual registers such as Roznamcha, FIR Register, Arrest Register, etc. with secure AI-managed digital platforms.
- Automatically updates entries based on officer actions, FIRs, arrests, and investigations.
- Detect inconsistencies or missing entries using anomaly detection.

5. AI Portal for Section 173 CrPC

- AI Portal require the Investigation Officers to upload daily investigation progress with regards to Challan under Section 173 of CrPC, so that it will ensure accountability of officers.
- AI Portal also contain the investigation record and the reasons for incomplete investigation, if so, in Interim Challan whereof the Challan has not been submitted within 14 days. The Portal will ensure transparency in investigation and fairness.

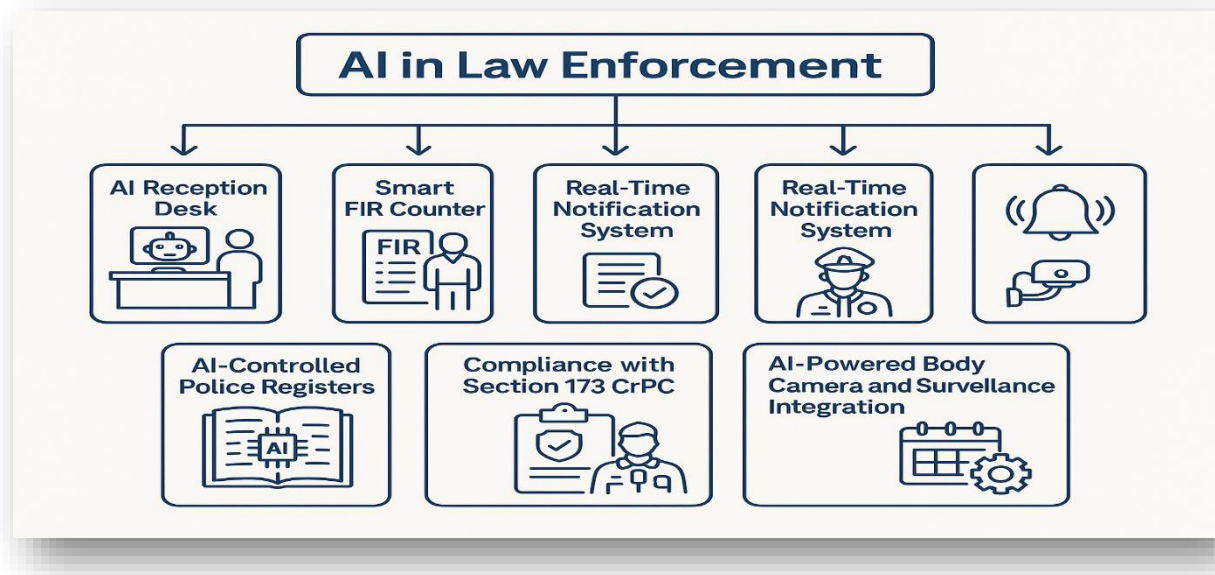
6. AI-Powered Body Camera and Surveillance Integration

- AI-powered body cameras monitoring officers' functioning, and easy commands by superior authorities or from control room.
- Detect use of force, verbal abuse, or violation of SOPs automatically.
- Real-time integration with CCTV and drone surveillance.

7. Automated Forensic Scheduling System

- AI Portal for Automated Forensic Scheduling System contain and maintain the schedule for forensic of collected evidence decreasing the burden and procedural complexities in forensic labs, and ensures transparency and accountability.

A graph showing the implementation of AI modules in law enforcement agencies.



Implementation Strategy and Pilot Model

Through following process and phases, the proposed policy will be implemented;

1. Implementation Phases

1.1. Phase I: Pilot Program (1 Year)

Phase I will be the initial implementation phase wherein the AI modules will be applied and implemented in a single police station of each province and the federal territory for one year.

The police station for the implementation of AI modules shall be determined by the Provincial Government for the province concerned, and by the Federal Government for Federal Territory, as the case may be.

The Police Department, in case of province, and the Ministry of Interior, in case of Federal Territory, shall execute, check, monitor, and supervise the implementation of AI modules.

1.2. Phase II: Expansion and Monitoring (2-3 Year)

In Phase II, the implementation plan will be expanded and monitored up to a single division of each province for 2 to 3 years.

The determination of the division for implementation will be as same as in Phase I.

The Provincial Government for the province concerned, and the Federal Government for the Federal Territory, as the case may be, shall make and provide adequate mechanism, fund, and infrastructure as required for such implementation.

1.3. Final Phase: Full-scale Rollout

In final phase, the AI modules will be applied and implemented to the law enforcement agencies of all provinces and the federal territory as whole.

The AI Governance Wing shall be established in each province and in federal territory to supervise and control the applications of AI in the agencies.

2. Infrastructure Development and Technological Requirements

2.1. Technology Infrastructure

2.1.1. High Speed Internet Connectivity

- Secure and stable internet connections with backup lines in all police stations and the control rooms.
- Internal Police Network (IPN) with end to end encryption.

2.1.2. Centralized Cloud System

- Deployment of a secure cloud-based data centre to host all AI applications, databases, and police registers.
- Data centres at division and provincial level.

2.1.3. Compliance with National Security Policies

- Integration of database centres with national IT Departments to comply with national security policies.

2.1.4. Integrated Software Programs

- Unified platforms integrating FIR registrations, case management, forensic scheduling, body camera feeds, and police registers.
- AI tools such as Natural Language Processing (NLP) module for Smart FIR Counter and Predictive Analytics for Real-time Notification System.

2.1.5. Mobile Application Support

- AI-powered mobile apps for police officers to access case data, receive real-time notifications, and manage body camera feeds remotely.

2.1.6. AI Reception Desk Integration

- Virtual Assistant Systems installed at entry points, integrated with main database for identity verification and task assignment.

2.2. Data Security

2.2.1. Cyber Security Audits

- Mandatory audits of AI systems, databases, and cloud infrastructure.
- Penetration testing to fix and identify vulnerabilities.

2.2.2. Data Backup and Disaster Recovery

- Automated daily backup of all AI system data.
- Disaster recovery plan to restore operations within 24 hours in case of any cyberattack or system failure.

2.2.3. Authorized Access Control Management

- Biometric authentication (Fingerprint/ Face ID) for officer's login to systems.

2.3. Hardware Requirements

2.3.1. AI Reception Desks:

- Touchscreens with facial recognition cameras and AI chat assistants.
- Noise-cancellation microphones for voice-based interaction.

2.3.2. Smart FIR Counters:

- High-end computer systems with speech-to-text capabilities and digital signature support.
- Document scanners and fingerprint readers for instant verification.

2.3.3. AI-Controlled Police Registers:

- Servers with AI processing capability.

- Secure, tamper-proof devices (tablets/laptops) for field officers.

2.3.4. Body Cameras and Surveillance:

- AI-integrated body cameras with real-time transmission ability.
- High-resolution CCTV systems integrated with AI for facial recognition, anomaly detection, and predictive behaviour analytics.

2.3.5. Forensic Scheduling Systems:

- Tablets or mobile devices for IOs (Investigating Officers) to manage forensic lab appointments and evidence submission.
- Integration with lab management software.

2.3.6. Command and Control Centres:

- Centralized monitoring rooms equipped with large displays showing live AI analytics, case progress reports, and officer deployment.

3. Training and Capacity Building

3.1. Police Officers

Police officers will receive training to operate AI-based systems like Smart FIR Counters, AI Reception Desks, and Body Cameras. They will be trained on data handling, privacy protocols, ethical use of AI, and real-time field applications. Regular refresher courses will keep them updated on system upgrades and legal changes.

3.2. Judicial Personnel

Judges, magistrates, and prosecutors will be trained to understand AI-generated evidence, digital reports, and forensic data management. They will learn to assess the authenticity of AI outputs and identify any biases or errors that could impact fair trials. Regular workshops will ensure ongoing familiarization with AI developments in law enforcement.

3.3. Technicians

Technical staff will be trained in installation, maintenance, cybersecurity, and troubleshooting of AI systems deployed in police stations. They will manage upgrades, conduct system audits, and ensure the smooth operation of AI modules. Annual certifications and upskilling programs will maintain technical excellence and system reliability.

4. Budget and Financial Considerations

4.1. Initial Investment

The initial investment will cover the setup of technology infrastructure, purchase of hardware such as AI screens, body cameras, servers, and the development of customized AI software. Costs will also include training programs for police officers, judicial personnel, and technicians to ensure system readiness.

4.2. Ongoing Costs

Ongoing costs will include system maintenance, software updates, cybersecurity management, cloud storage fees, and periodic training refreshers. Additional budget will be provided for technical support, hardware replacements, and electricity and internet charges for continuous operations.

4.3. Public-Private Partnerships

To manage financial pressure, partnerships with private technology firms, and telecom companies, can be developed. These partnerships can assist in funding AI infrastructure, sharing technical expertise, and ensuring innovation while reducing direct government expenditure.

5. Monitoring and Evaluation Framework

5.1. Performance Indicators

Specific and measurable performance indicators will be developed to assess the efficiency and effectiveness of AI systems. These will include the number of FIRs registered through Smart FIR Counters, reduction in case processing times, compliance rate with Section 173 CrPC, uptime of AI reception desks, accuracy of AI surveillance alerts, and officer satisfaction scores. Regular monitoring of these indicators will help track progress and highlight areas needing improvement.

5.2. Feedback Mechanisms

Continuous feedback loops will be established involving both internal users (police officers, technicians) and external stakeholders (citizens, judicial officers). Surveys, interviews, and digital feedback forums will be used to gather opinions on system usability, transparency, and areas of concern. Special feedback points will also be integrated into AI systems like the reception desk screens to record public input after services are rendered.

5.3. Impact Assessment

A comprehensive impact assessment will be conducted annually to evaluate how AI adoption has improved policing outcomes. This will include measuring

reduction in crime rates, enhancement in public trust, faster case resolution, increased transparency, and judicial reliance on AI-generated evidence. Independent external evaluators or academic institutions will be engaged to ensure impartial and credible reporting of the results.

6. Risk Management and Contingency Plans

6.1. Risks

Several risks are associated with the implementation of AI in police stations, which must be proactively managed. Technical failures, such as system crashes, server downtimes, or malfunctioning AI algorithms, could disrupt operations. Cybersecurity threats, including hacking, data breaches, and ransomware attacks, pose a serious risk to sensitive police information. There is also the potential for bias or errors in AI decision-making, which could lead to wrongful actions or undermine public trust. Resistance to change from police personnel, lack of adequate training, and dependency on technology without human oversight further increase operational risks. Additionally, legal and ethical risks may arise if AI-generated evidence is challenged in courts or if citizen privacy rights are compromised.

6.2. Contingency Plans

To address technical failures, a robust backup system including secondary servers and offline manual operation protocols will be established to ensure uninterrupted services. Comprehensive cybersecurity measures such as regular audits, firewalls, encryption, and penetration testing will be maintained, alongside an emergency response team to manage cyberattacks. AI outputs will always be reviewed by human officers to mitigate bias and errors, ensuring accountability remains with trained personnel. Change management programs, including continuous training and motivation strategies, will be implemented to overcome staff resistance and build confidence in new technologies. Legal advisory teams will be set up to review all AI applications to ensure compliance with constitutional protections and international human rights standards. Regular drills and simulations will prepare the police stations to respond effectively to system failures, cyberattacks, or legal challenges, ensuring resilience and sustained service delivery.

Integration with the Criminal Justice System

1. Real-Time Connectivity with Judicial Magistrates

The integration of AI-governed police stations with judicial magistrates will revolutionize the speed, transparency, and fairness of the criminal justice process. Through secure, real-time digital connectivity, police stations will be able to submit FIRs, investigation reports, evidence lists, and challans (under Section 173 CrPC) directly to the magistrate's digital platform without physical delays. Judicial magistrates will have immediate access to updated case files, allowing them to issue orders, warrants, and approvals electronically within minutes rather than days.

AI modules will assist in automatically formatting and organizing police submissions into legally compliant documents, ensuring that magistrates receive complete and properly structured case files. Magistrates will also be able to send digital queries or observations back to the police through the system for quick compliance, reducing unnecessary adjournments and paperwork. Real-time tracking of court orders and their implementation will enhance accountability on the part of police officers.

This seamless linkage will ensure better protection of the rights of accused persons, faster decision-making on bail or remand applications, and improved compliance with time-bound legal requirements. Ultimately, the real-time integration between AI-policed stations and the judiciary will bring about a significant reduction in case backlogs, help in timely delivery of justice, and build public trust in the criminal justice system.

2. Integration with Public Prosecution Departments

The integration of AI-governed police stations with Public Prosecution Departments will ensure that the process of case preparation, scrutiny, and trial management becomes faster, more accurate, and more transparent. Once an FIR is registered and investigation progresses, AI systems will automatically compile investigation reports, evidence summaries, witness lists, and forensic reports in a standardized digital format accessible to public prosecutors in real time.

Public prosecutors will be able to review investigation files immediately through secure digital platforms, allowing them to guide police officers proactively on evidence collection, legal requirements, and case-building strategies. This early and consistent

coordination will improve the quality of investigations and reduce the chances of weak or flawed cases reaching the courts. Prosecutors will also be able to electronically request additional documents, clarifications, or further investigations through the AI system, ensuring a continuous flow of communication.

The AI modules can generate alerts for prosecutors regarding deadlines under criminal law, like submission of challans or expiry of remand periods, promoting strict legal compliance. In addition, smart analytics tools can assist prosecutors in identifying patterns of recurring procedural mistakes, helping police departments to improve their performance through targeted training and policy adjustments.

Overall, integration with Public Prosecution Departments will minimize delays, strengthen case quality, increase conviction rates, and help in delivering faster and fairer justice to society.

3. Collaboration with Forensic Science Agencies

Collaboration between AI-governed police stations and forensic science agencies will create a faster, more reliable, and scientifically sound investigation process. AI modules at police stations will be linked directly to forensic laboratories through secure digital channels, allowing immediate scheduling of forensic tests, real-time tracking of evidence status, and instant sharing of preliminary forensic findings.

When physical evidence like fingerprints, DNA samples, ballistic materials, or digital devices is collected, AI systems will generate standardized forensic request forms and electronically notify the concerned forensic labs. This will eliminate traditional delays caused by manual paperwork and transport bottlenecks. Automated reminders and alerts will ensure that forensic examinations are conducted within legally required timelines, aiding in faster case progression.

AI integration will also enable forensic labs to send digital forensic reports directly back to the investigating officers and prosecutors, reducing the risk of report tampering, loss, or miscommunication. Smart analytics within the system can help prioritize urgent cases, such as murder or terrorism investigations, ensuring optimal utilization of forensic resources.

Through this collaboration, the quality of scientific evidence presented in court will significantly improve, investigations will become more credible, and the entire criminal justice process will be strengthened through accuracy, speed, and transparency.

Legal, Ethical, and Practical Considerations

The implementation of AI modules in criminal justice system and in law enforcement agencies may raise some legal, ethical, and practical concerns which should be kept in mind while applying those AI modules. To tackle those concerns, some better solutions are to be given. Following are the points describing the concerns and their safeguards in implementation of AI modules;

1. Ethical Concerns and Safeguards

1.1. Algorithmic Bias and Fairness

AI systems may unintentionally replicate or amplify existing biases in policing data, leading to unfair targeting or discrimination against certain groups, especially minorities or marginalized communities. To counter this, AI algorithms should undergo regular bias audits conducted by independent experts. Diverse and representative datasets must be used during AI training to minimize skewed results. Additionally, fairness standards must be legally enforced, requiring all AI models used in policing to meet clear non-discrimination and equity benchmarks.

1.2. Right to Human Oversight and Appeal

There is a risk that decisions made by AI, such as recommendations for arrest or investigation prioritization, might be accepted blindly without human review, potentially leading to unjust outcomes without recourse for correction.

A mandatory human oversight policy should be implemented, ensuring that all AI-generated recommendations are reviewed and approved by trained police officers or judicial personnel. Citizens must also be provided with a clear right to appeal any action based on AI assessments, ensuring that human rights and procedural fairness are protected at every stage.

1.3. Establishment of AI Ethics Oversight Board

Without a dedicated body to monitor ethical use, AI deployment in law enforcement could drift toward misuse, lack of transparency, and erosion of public trust. An independent AI Ethics Oversight Board should be created, comprising legal experts, human rights activists, technologists, police leadership, and members of civil society. This Board will monitor AI usage, review complaints, conduct ethical impact assessments, and publish annual

transparency reports. It will also have the authority to halt or suspend AI tools that violate ethical standards.

2. Data Privacy and Cybersecurity

2.1. Need for a Comprehensive Data Protection Law

The use of AI modules in police work will involve handling vast amounts of sensitive personal data, including biometric records, criminal histories, and private communications. Without a strong legal framework, there is a serious risk of misuse, unauthorized access, or violations of citizens' privacy rights.

A comprehensive national Data Protection Law must be enacted, specifically tailored to address the unique challenges posed by AI use in law enforcement. This law should set strict standards for data collection, storage, sharing, and deletion. It must guarantee individuals' rights to know what data is collected about them, seek corrections, and demand deletion where appropriate. Clear provisions for consent, purpose limitation, and oversight by an independent Data Protection Authority should be embedded to ensure transparency and accountability.

2.2. Technical Safeguards and Penalties

AI systems are vulnerable to cyberattacks, hacking, unauthorized surveillance, and internal data breaches, which could compromise sensitive criminal justice information and public trust. Strong technical protections must be mandatory, including end-to-end encryption, multi-factor authentication, regular vulnerability assessments, and intrusion detection systems for all AI-related platforms. Data access must be strictly role-based, ensuring only authorized personnel can view or modify sensitive records. Strict criminal penalties, including fines, suspension, and imprisonment, must be established for any individual or entity found guilty of unauthorized access, data manipulation, or cyberattacks. Regular cybersecurity training for all police personnel will further strengthen the defence against internal and external threats.

3. Constitutional and Legal Compatibility

3.1. Compliance with Fundamental Rights

The deployment of AI in police stations must not infringe upon the fundamental rights guaranteed under the Constitution of Pakistan, such as the right to due process, privacy, dignity, equality, and protection against arbitrary arrest and detention.

Every AI system must be designed and operated strictly in accordance with constitutional rights. Before implementation, a thorough constitutional review must be conducted to ensure that AI practices do not lead to racial profiling, arbitrary decision-making, or violation of personal freedoms. AI actions affecting individuals should be transparent, subject to human verification, and accompanied by legal remedies in case of rights violations. Specific protocols should be established to ensure that any data collection, surveillance, or decision-making respects the principles of legality, necessity, and proportionality.

3.2. Legislative Amendments for AI Recognition

Current Pakistani laws, including the Code of Criminal Procedure (CrPC), Police Rules, and Evidence laws, do not recognize or regulate the role of AI systems, which may create legal uncertainty about the validity of AI-generated evidence, reports, or decisions. Targeted legislative amendments must be introduced to formally recognize the role of AI in law enforcement activities. Provisions should define the scope, admissibility, and legal status of AI-generated documents, evidence, and recommendations. Rules must be established to govern AI's operational boundaries, ensuring that human oversight remains paramount. New legal standards should also address accountability mechanisms for errors or abuses arising from AI use, thus embedding AI within the criminal justice framework with legal certainty and public legitimacy.

Final Remarks

In an era where artificial intelligence is rapidly transforming societies, it is imperative for Pakistan to establish a balanced framework that embraces innovation while safeguarding fundamental rights. This policy draft aims to provide a comprehensive approach by proposing legislative measures and strategic guidelines for the ethical use of AI, with particular focus on facial recognition, data protection, and fingerprint technologies. By adopting these reforms,

Pakistan can not only ensure responsible AI development but also build public trust in emerging technologies.

Through collaboration between government institutions, the private sector, academia, and civil society, we can foster an AI ecosystem that is inclusive, transparent, and resilient. The proposed policy emphasizes the need for strong regulatory oversight, protection of personal data, and alignment with international best practices, while being rooted in our national context. It is hoped that with proactive implementation, Pakistan will position itself as a regional leader in ethical AI governance, ensuring that technological advancement translates into broader societal benefit.

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